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Studierichting: Informatica
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$$z^3 = 8 \Rightarrow z^3 = 8 + 0i$$

$$r = \sqrt{0^2 + 8^2} = 8$$

$$\tan(\theta) = \frac{0}{8} = 0 \Rightarrow \theta = \text{Bgtan}(0) = \pi$$

$$z^3 = \sqrt[n]{r} \operatorname{cis} \left(\frac{\theta + 2k\pi}{n} \right)$$

$$\Rightarrow \sqrt[3]{8} \operatorname{cis} \left(\frac{0 + 2k\pi}{3} \right)$$

$$z_0 = 2 \operatorname{cis}(0) = 2(\cos(0) + i \sin(0)) = 2(1 + 0i) = 2$$

$$z_1 = 2 \operatorname{cis} \left(\frac{2\pi}{3} \right) = 2 \left(\cos \left(\frac{2\pi}{3} \right) + i \sin \left(\frac{2\pi}{3} \right) \right)$$

$$= 2 \left(\frac{-1}{2} + \frac{\sqrt{3}}{2}i \right) = -1 + i\sqrt{3}$$

$$z_2 = 2 \operatorname{cis} \left(\frac{4\pi}{3} \right) = 2 \left(\frac{-1}{2} + \left(\frac{-\sqrt{3}}{2} \right) i \right) = -1 - i\sqrt{3}$$

References